

title: F · Delivery Plan, Infrastructure & Cost Model description: A grounded, bottom-up cost and delivery model for the DepEd HRIS — 2026 PH-market rates for the build team, on-premises infrastructure sizing with a public-cloud alternative, month-by-month staffing curve, and a full breakdown against the PHP 500 M ABC. Nothing here is a bidder quote; every number carries its assumptions.

F · Delivery Plan, Infrastructure & Cost Model

!!! info “How this paper is different” Papers **A / B / C** describe **what** the PBD requires and **how** a bidder could respond. Paper **D** describes the **value-adds** that beat competitors at no incremental cost. Paper **E** shows the **international precedent**.

****This paper answers a different question:**** *is the ABC of PHP 500 M realistic for what the PBD
This is a ****modelling exercise****, not a bid. Actual bids will differ by team blend, subcontractor

F.1 Guiding assumptions

Before any peso is committed, six assumptions have to be true. Change any of them and the cost model shifts materially.

- 1. Two deployment options priced — DepEd (or the bidder) selects at inception.** **Option A** is on-premises in DepEd-owned facilities (Central Office + a DR site). **Option B** is public cloud on GovCloud PH or an in-country region of AWS Manila / Azure PH. Both are fully compliant with RA 10173 data-residency requirements when configured per [Paper I §1.5](#). This paper prices, sizes, and compares both — pros, cons, and 24-month cost. The single-option recommendation is deferred to inception (M1), where local DC readiness, DICT policy, FX appetite, and 5-year TCO can be weighed against actual conditions. **Both fit inside the PHP 500 M ABC.**
- 2. Modular monolith at go-live, extracted to services only where the load profile demands.** Payroll gets its own service first (M6 gate), everything else stays modular within a common code base. Rationale: reduces the surface area of change during a 365-day build, and matches the working-set the internal team can hire against at PH rates.
- 3. Team is Philippines-based, senior-heavy, and blended in-house + subcontract-free.** SCC clause 7 forbids subcontracting, so consulting-firm augmentation is explicitly excluded. This constrains the team to what the prime contractor can staff directly.
- 4. Rates are 2026 PH market medians** taken from JobStreet PH, Kalibrr, and the WTW Philippines IT compensation report (median values, private-sector, 5-yr-experience tier where applicable). Rates are **fully-loaded** (base × 1.35 to cover 13th month, allowances, HMO, SSS/PhilHealth/Pag-IBIG employer share, retirement fund, and equipment). Consulting-firm billable rates would be 1.8×–2.5× these numbers — outside the ABC without major scope cuts.

5. **12-month build, 12-month warranty.** The PBD contract runs 365 days from NTP to final acceptance (M8). A conservative bid includes a **12-month post-acceptance warranty** priced as a separate line item; unpriced warranties are how bidders lose margin in year 2.
6. **PHP:USD 55.50** is used for any USD-denominated cloud or software list-price conversion. This is the 12-month trailing median rate at time of writing. Contracts should hedge or specify a currency escalation clause.

F.2 The scope, priced in one paragraph

The PBD demands a **1 M+ employee HRIS**, delivered in **365 days**, with **11 functional modules**, **99% uptime**, four SLA response bands, integration with **9 external regulators**, **28 recruitment reports**, offline access for schools, and full source-code turnover at M8. That is roughly the same envelope as **Malaysia's HRMIS** (1.6 M civil servants) or **NHS ESR** (1.5 M staff) — systems that took **3–5 years and 2–4× the ABC** to reach their current form. The Philippine ABC is priced for a **compressed first-generation delivery** that will need iterative hardening in years 2–3 through a separate maintenance contract. Any bidder who prices as if 365 days is enough for a mature system will lose money in year 2.

F.3 Team composition and monthly cost

F.3.1 Reference rates — 2026 PH market, fully loaded

Rates below are **PHP per month, fully-loaded** (base × 1.35 factor for 13th month + statutory contributions + HMO + equipment). Sources: WTW Philippines IT Comp 2025 (extrapolated +5% for 2026), JobStreet PH 2025 salary reports, Kalibrr median postings Q4 2025 – Q1 2026. Median values shown; a bidder building slack should add 10–15%.

Role	Level	Monthly base (PHP)	Fully loaded (× 1.35)
Program Manager	15+ yr	220,000	297,000
Delivery Manager / PMP	10+ yr	170,000	229,500
Solutions / Enterprise Architect	12+ yr	230,000	310,500
Tech Lead (per bounded context)	10+ yr	180,000	243,000
Senior Software Engineer	7+ yr	140,000	189,000
Software Engineer	4–7 yr	90,000	121,500
Junior Software Engineer	1–3 yr	55,000	74,250
DevOps / SRE	7+ yr	155,000	209,250
Data Engineer	5+ yr	120,000	162,000
Security Engineer / DPO liaison	8+ yr	190,000	256,500
Database Engineer (PostgreSQL)	8+ yr	165,000	222,750
QA Lead	8+ yr	130,000	175,500
QA Engineer	3–6 yr	75,000	101,250
Business Analyst / Module Owner	6+ yr	105,000	141,750
UI / UX Designer	5+ yr	95,000	128,250
Technical Writer	4+ yr	65,000	87,750
Change Management / Trainer	5+ yr	80,000	108,000
Migration Specialist	6+ yr	115,000	155,250

!!! note “Why loaded rates, not billable rates” A consulting firm billing DepEd on a T&M basis would apply a 2.0×–2.5× multiplier to cover overhead, bench, sales, and margin. Consulting-firm rates would push a comparable team to **PHP 15–20 M / month**, which cannot fit inside a 12-month PHP 500 M ABC. This model assumes the prime contractor **employs the team directly** (no subcontract per SCC 7) and charges margin at the contract level, not the resource level.

F.3.2 Team composition — peak headcount (M4–M9)

Function	Roles	Count	Loaded / mo (PHP)
Program leadership	Program Manager × 1, Delivery Manager × 2	3	756,000
Architecture	Solutions Arch × 1, Enterprise Arch × 1	2	621,000
Engineering leadership	Tech Lead × 3 (Core HR, Payroll, Recruitment)	3	729,000
Senior engineering	Senior SWE × 10	10	1,890,000
Mid-level engineering	SWE × 14	14	1,701,000
Junior engineering	Jr SWE × 6	6	445,500
Platform / SRE	DevOps × 4	4	837,000
Data	Data Engineer × 2, DB Engineer × 2	4	769,500
Security	Security Eng × 2	2	513,000
QA	QA Lead × 2, QA Eng × 8	10	1,161,000
Product / BA	Business Analyst × 4	4	567,000
Design	UI/UX × 3	3	384,750
Documentation	Tech Writer × 2	2	175,500
Adoption	Trainer × 4	4	432,000
Migration	Migration Specialist × 2	2	310,500
Peak headcount		69	PHP 11,292,750 / month

Peak monthly personnel = PHP 11.3 M. This is the ceiling. Actual monthly cost tracks the staffing curve in §F.4 and averages materially less.

F.3.3 Team composition — steady-state (warranty year, M13–M24)

Post-acceptance the team shrinks to a **support-shaped** shape: fewer builders, more operators, all runbook-driven.

Function	Count	Loaded / mo (PHP)
Delivery Manager	1	229,500
Solutions Architect (part time, 50%)	0.5	155,250
Tech Lead	1	243,000
Senior SWE	3	567,000
Mid SWE	4	486,000
DevOps / SRE	3	627,750
DB Engineer	1	222,750
Security Eng	1	256,500
QA	2	202,500
Support / Service Desk lead	1	141,750
Warranty team	17.5	PHP 3,131,000 / month

Warranty year total = 12 × PHP 3.13 M = PHP 37.6 M.

F.4 Staffing curve — monthly headcount and cost

The team is not level-loaded. It ramps from 8 FTE at kick-off to 69 at peak (M4–M9), then tapers as UAT completes and the warranty team takes over.

Month	Milestone context	FTE	Personnel spend (PHP)
M1 (mo 1)	Inception, discovery, hiring	12	1,900,000
Mo 2	Design ramp	28	4,600,000
M2 (mo 3)	Architecture baseline	42	7,000,000
Mo 4	Core HR + ESS build starts	58	9,600,000
M3 (mo 5)	Peak — Core HR + ESS	69	11,300,000
Mo 6	Recruitment + Notifications build	69	11,300,000
M4 (mo 7)	Peak — recruitment done	69	11,300,000
Mo 8	Payroll build	65	10,700,000
M5 (mo 9)	Payroll build complete	62	10,200,000
Mo 10	Parallel-run + hardening	55	9,100,000
M6 (mo 11)	Parallel-run gate (critical)	50	8,300,000
M7 (mo 12)	UAT + training + go-live	40	6,700,000
Build total			PHP 102,000,000
Mo 13 (warranty starts)	Hyper-care, month 1	25	4,600,000
Mo 14–15	Hyper-care taper	22, 20	3,900,000 + 3,500,000
Mo 16–24	Warranty steady-state	17.5 avg	3,131,000 × 9 = 28,180,000
Warranty total			PHP 40,180,000

Personnel total (build + warranty) = PHP 142.2 M.

The curve is drawn as a Mermaid chart in §F.11.

F.5 Deployment Option A — On-premises

Primary DC at DepEd Central Office (Malugay St., Makati). **DR site** at a DICT-designated GovCloud PH facility or contracted colocation. This option is chosen when data residency is a hard control, upfront capital is available, 5-year TCO is prioritised over month-1 spend, and DepEd's DC has space / power / cooling / staff capacity confirmed by M1.

F.5.1 Sizing model — the numbers this is built on

A 1 M+ employee HRIS is not the same as a 1 M concurrent-user system. The concurrent working-set is roughly:

Load profile	Concurrency	Trigger
Ordinary weekday	~ 8,000	Employees checking leave balances, HRO doing appointments
Payroll-cutoff day	~ 45,000	Payroll officers + employees verifying payslips
Recruitment-deadline day	~ 55,000	External applicants + Recruitment Officers
CSC-form-9 batch day	~ 12,000	Job postings + notifications
Sized ceiling (design point)	80,000	1.5× the historical peak of legacy systems

Design point: **80,000 concurrent users** with a **P95 latency budget of 3 seconds** under normal load (PBD requirement, [Paper A §A.3](#)). At 80 K × 5 req/user/minute the API-tier peak is roughly **6,700 req/s** — well within a modest Kubernetes cluster's capacity.

F.5.2 Compute — production environment

Primary DC at DepEd Central Office. Second site (DR) at a DICT-designated GovCloud PH facility or contracted colocation.

Component	Spec	Units	Purpose	List CAPEX (PHP)
Kubernetes master	8 vCPU / 32 GB / 480 GB SSD	3	Control plane, HA	PHP 1,500,000
Kubernetes worker	32 vCPU / 128 GB / 1 TB NVMe	12	Application tier	PHP 12,000,000
PostgreSQL 16 primary	32 vCPU / 256 GB / 4 TB NVMe RAID10	1	Core HR + Payroll + Recruitment OLTP	PHP 2,800,000
PostgreSQL 16 replica	Same spec	2	Sync replica (HA) + async (reporting)	PHP 5,600,000
Redis cluster	8 vCPU / 64 GB	3	Session + cache + rate limiting	PHP 2,100,000
OpenSearch cluster	16 vCPU / 64 GB / 2 TB SSD	3	Search + audit log + retrieval index (D \$3)	PHP 4,500,000
Object storage (Ceph / MinIO)	8 vCPU / 64 GB / 12 × 20 TB HDD	4	201 files, attachments — 240 TB raw / 80 TB usable	PHP 6,400,000
BPMN workflow engine host	16 vCPU / 64 GB	2	Approvals + recruitment stages	PHP 2,600,000
Load balancer / WAF appliance	HA pair	2	Ingress + DDoS + rule engine	PHP 3,800,000
Firewall (next-gen)	HA pair	2	Perimeter + segmentation	PHP 4,200,000
Backup server	16 vCPU / 128 GB / 200 TB dedup	1	Veeam or Bacula target	PHP 3,500,000
Monitoring stack host	16 vCPU / 128 GB / 4 TB SSD	2	Prometheus + Grafana + Loki + Tempo	PHP 1,700,000
Network core (spine + leaf)	10G / 25G fabric		Rack fabric + patch	PHP 3,600,000
Rack, power, UPS, cabling	3 racks fully populated		Physical infrastructure	PHP 2,500,000
Setup, racking, commissioning	Once		Vendor engineering	PHP 1,500,000
Production DC subtotal				PHP 58,300,000

F.5.3 DR site — cost-reduced hot standby

DR runs at ~40% capacity, warm-standby for OLTP (streaming replication) and cold for the analytical replica. Full failover target: 4-hour RTO, 15-minute RPO.

Component	DR sizing	List CAPEX (PHP)
K8s cluster (small)	3 masters + 6 workers	PHP 7,500,000
PostgreSQL standby	Primary spec	PHP 2,800,000
Redis (small)	2 nodes	PHP 1,400,000
OpenSearch (small)	2 nodes	PHP 2,000,000
Object storage	100 TB raw	PHP 3,200,000
Load balancer / WAF	Single unit	PHP 1,900,000
Firewall	HA pair	PHP 4,200,000
Network + rack + setup		PHP 2,800,000
DR subtotal		PHP 25,800,000

F.5.4 Non-production environments

Four non-prod environments run in the primary DC on a dedicated smaller cluster (or virtualised on the DR hardware during quiet windows):

Environment	Purpose	Sizing	CAPEX (PHP)
DEV	Developer sandboxes	4 workers × 16 vCPU	PHP 2,400,000
SIT	System integration	Full stack, 20% of prod	PHP 3,800,000
UAT	User acceptance	Full stack, 40% of prod	PHP 6,200,000
Training	Parallel training env	Full stack, 30% of prod	PHP 4,500,000
Non-prod subtotal			PHP 16,900,000

F.5.5 Infrastructure total — on-prem, one-time CAPEX

Bucket	PHP
Production DC	58,300,000
DR site	25,800,000
Non-production (DEV / SIT / UAT / Training)	16,900,000
On-prem CAPEX total	PHP 101,000,000

F.5.6 Recurring OPEX — power, bandwidth, colo, warranties (24 months)

Item	Monthly (PHP)	24-mo total (PHP)
Power + cooling (primary + DR)	320,000	7,680,000
Bandwidth (2 × 1 Gbps DIA + PhilSys/GSIS peering)	280,000	6,720,000
Colocation for DR	190,000	4,560,000
Hardware warranty (3-yr next-business-day)	450,000	10,800,000
CloudFlare or DDoS provider	90,000	2,160,000
Cert authority + code signing	25,000	600,000
OPEX subtotal (24 mo)		PHP 32,520,000

F.6 Deployment Option B — Public cloud

If DepEd instead deploys to **GovCloud PH** (DICT-managed sovereign cloud) or a hyperscaler with data residency in-country, the CAPEX collapses into OPEX. This option is priced for AWS Manila / Singapore with an ap-southeast-1 primary and a warm DR in ap-southeast-3 (Jakarta), or the equivalent on GovCloud PH via DICT. This option is chosen when speed of provisioning outweighs capital planning, managed services (RDS, ElastiCache, OpenSearch) meaningfully reduce ops burden, elastic scaling is preferred to over-provisioning, or DC readiness is uncertain.

F.6.1 Reserved-instance model, 24-month term

Component	Instance class	Monthly (PHP)
EKS control plane	Managed	42,000
Application node group	12 × c6i.8xlarge (32 vCPU, 64 GB), 3-yr RI	780,000
RDS PostgreSQL	db.r6g.8xlarge Multi-AZ, primary + replica	620,000
ElastiCache Redis	3 × cache.r6g.2xlarge	165,000
OpenSearch	3 × r6g.4xlarge.search, 2 TB gp3	285,000
S3	80 TB standard + 40 TB glacier	110,000
CloudFront + WAF + Shield Advanced		195,000
NAT gateways + data transfer		145,000
Route 53 + KMS + Secrets Manager		55,000
CloudWatch + X-Ray		90,000
Backup vault (cross-region)		75,000
DR (warm standby, ~40%)	Duplicated at ap-southeast-3	620,000
Non-prod (DEV + SIT + UAT + Training)	~40% of prod	780,000
Monthly total		PHP 3,962,000
24 months		PHP 95,088,000

F.6.2 Cost comparison over 24 months

	Option A · On-prem (\$F.5)	Option B · Cloud (\$F.6.1)
Capital investment	PHP 101,000,000	0
Operating cost 24 mo	32,520,000	95,088,000
Total 24 mo	PHP 133,520,000	PHP 95,088,000
Residual value at 24 mo	~ PHP 45 M (hardware, 3–5 yr useful life)	0
Net 24-mo cost of ownership	~ PHP 88,500,000	PHP 95,088,000

Two ways to read the same numbers:

- **Over 24 months only** — cloud is ~PHP 6.6 M more expensive if the on-prem hardware is fully written off in that window. Meaningless because hardware has 5-yr useful life.
- **Amortised over 5 years** — on-prem hardware drops to ~PHP 20 M/yr, making **on-prem cheaper from month 30 onwards**. Cloud OPEX stays flat at ~PHP 47.5 M/yr.

Neither number is a recommendation. Both are inputs to the decision framework in §F.6.4.

F.6.3 Pros and cons

Option A · On-premises

✓ Pros	✗ Cons
Data residency by construction — data never leaves DepEd-owned premises	High upfront CAPEX — PHP 101 M concentrated in month 1–2
No FX exposure — all PHP-denominated hardware and warranties	Long lead time — 90–120 days to procure and rack hardware; procurement compresses M1
Physical control assurance for COA audit and DPO oversight	Requires DepEd DC readiness — space, power, cooling, staff must be confirmed by M1
Hardware amortises to ~PHP 20 M/yr from year 3 (5-yr useful life) — cheaper long-term	Slower capacity expansion — need to buy hardware for scale-up
No cloud egress or bandwidth charges in the operating model	Hardware refresh cycle every 5 years — a separate capital ask
Predictable cost line — CAPEX is one-time, OPEX is fixed	Requires in-house SRE talent to operate 24×7 (~4 FTE minimum)
Full customisation of network topology — VLANs, firewalls, air-gaps	DR-site provisioning takes 6–9 months — cannot go live in DR before that
Simplest compliance narrative — physical assurance for RA 10173, COA, NPC	Vendor lock-in for hardware warranties — 3-yr commitments per vendor

Option B · Public cloud

✓ Pros	✗ Cons
Zero CAPEX — converts entirely to OPEX; no capital ask upfront	Higher OPEX over 24 months — PHP 95 M vs PHP 32.5 M for on-prem
Elastic scaling — payroll-cutoff bursts scale automatically; no over-provisioning	FX exposure if not on GovCloud PH — AWS / Azure prices are USD-denominated
Rapid provisioning — new environments in hours, not months	Data residency requires strict region pinning — every service must be checked
Managed services (RDS, ElastiCache, OpenSearch) reduce ops burden	Cross-over point where on-prem becomes cheaper: month 30 (5-yr TCO)
Built-in multi-AZ DR at lower cost than physical DR	Cloud egress and inter-region transfer charges add up quickly
Continuous security updates handled by provider	COA audit trail more complex when infra is a shared-tenant service
No hardware refresh cycle — infrastructure is always current	Cloud-vendor API lock-in (KMS, IAM, managed services) needs mitigation
Smaller ops team — 3 FTE cloud-native SRE vs 4+ on-prem	Requires FinOps discipline — cost predictability depends on RI + budget alerts
Faster non-prod provisioning — DEV / SIT / UAT / Training envs spun in hours	Compliance narrative more layered — must attest provider SOC 2 + ISO 27001

F.6.4 Decision framework — six questions to answer at inception

The choice between A and B should be made at M1, not in the bid, and should be documented in the inception report. Six questions drive the decision:

#	Question	Answer favours Option A	Answer favours Option B
1	Is DepEd DC readiness (space, power, cooling, staff) confirmed by end of M1?	Yes	No / uncertain
2	Is upfront CAPEX of PHP 101 M available in year 1?	Yes	No / prefer OPEX
3	Is FX exposure acceptable? (Only if on GovCloud PH is fully in scope: cloud can be FX-neutral)	Prefer none	Acceptable / hedged
4	Is the 5-year horizon more important than the 2-year contract window?	Yes	Only 2 years matter
5	Is the ops team hire-able at target scale (4+ FTE SRE + DBA)?	Yes	Prefer smaller cloud-native team
6	Do payroll and recruitment peaks demand elastic scaling?	No (steady load)	Yes (2–5× peak)

Rule of thumb. Four or more answers in one column → that option. A mixed profile means a **hybrid** (Option A primary + Option B DR, or the reverse) is worth pricing as a third scenario at inception.

F.6.5 Both options are inside the ABC

The full cost breakdown for each option is in \$F.9. Preview:

	Option A · On-prem	Option B · Cloud
Total 24-month contract price	PHP 421.4 M	PHP 358.5 M
% of ABC (PHP 500 M)	84.3%	71.7%
Bid headroom	PHP 78.6 M (15.7%)	PHP 141.5 M (28.3%)

Cloud is cheaper over the contract window; on-prem is cheaper from month 30 onwards. Both leave meaningful bid headroom for competitive positioning or additional value-adds.

F.7 Software, licensing, and third-party services

Nearly the entire stack is open source. Paid items are minimal but concentrated on **security, monitoring, and integration middleware**.

Item	Purpose	Cost (PHP)	Notes
Keycloak (FOSS)	OAuth2/OIDC, MFA, RBAC	0	Self-hosted
PostgreSQL 16 (FOSS)	OLTP	0	Community edition
Redis (FOSS)	Cache	0	Redis 7 with source-available note; consider Valkey
OpenSearch (FOSS)	Search + audit	0	Community
Kubernetes (FOSS)	Orchestration	0	Vanilla or K3s
Prometheus + Grafana	Monitoring	0	OSS core
RHEL 9 subscriptions	Server OS (30 sockets)	4,800,000 (2 yr)	₱80 k / socket / yr
Veeam Backup & Replication	Backup software	3,600,000 (2 yr)	Enterprise Plus
F5 or FortiWeb WAF licenses	WAF rulesets	2,400,000 (2 yr)	Included with appliance in some SKUs
Palo Alto NGFW licenses	Threat prevention + URL filtering	2,800,000 (2 yr)	Or Fortinet FortiGate
APM (Datadog or self-hosted SigNoz)	Traces + logs	2,400,000 (2 yr)	Self-hosted variant cuts this to ~600 k
Static analysis (SonarQube Enterprise)	Code quality gate	900,000 (2 yr)	Community edition is free but no CI enforcement
DAST (OWASP ZAP + Burp Enterprise)	Security testing	700,000 (2 yr)	
Load-testing SaaS (k6 Cloud or Grafana Cloud k6)	Perf runs at 80 K concurrent	500,000 (build only)	Peaks in M5 and M7
PhilSys eKYC API fees	Per-verification	800,000 (build)	Pass-through, one-off during migration
SMS gateway (D ₱1)	100 K messages/mo	1,800,000 (24 mo)	Globe / Smart enterprise SMS
Code-signing certificate	Windows / macOS agents	120,000 (24 mo)	
Software subtotal		PHP 20,820,000	

F.8 Training, change management, documentation

DepEd's 1 M+ employees do not need bespoke training — the ESS is designed for self-service, and only about **35,000 users** (HR officers, Payroll officers, Recruitment officers, Admin) need direct role-based training. The rest are reached via train-the-trainer and video modules.

Item	Cost (PHP)	Notes
Content development (video + LMS modules for ESS)	3,200,000	60 modules × 15 min
Role-based training curriculum (11 modules × 4 role types)	2,400,000	Materials + facilitator packs
Train-the-trainer sessions (17 regions × 2 rounds)	4,600,000	Regional trainer costs, materials, T&L reimbursement
Direct role-based training (35 K users, batched)	5,800,000	Venue, materials, connectivity — over 4 months
e-Learning platform integration	1,200,000	LMS content packaging (SCORM / xAPI)
User manuals + admin runbooks + API docs	1,700,000	350 pages, publishing
Video production (walkthroughs, quick-refs)	900,000	30 videos, professional edit
Change-management comms (posters, digests)	700,000	12-month campaign
Training + comms subtotal	PHP 20,500,000	

F.9 Cost model — full bottom-up

F.9.1 Build phase — Option A (on-prem, M1–M8, 12 months)

Bucket	PHP	% of build
Personnel (build, \$F.4)	102,000,000	30.0%
Infrastructure CAPEX (on-prem, \$F.5.5)	101,000,000	29.7%
Infrastructure OPEX (build 12 mo, \$F.5.6 pro-rata)	16,260,000	4.8%
Software + licensing (build 12 mo, \$F.7 pro-rata)	10,410,000	3.1%
Training + change mgmt (\$F.8)	20,500,000	6.0%
Subtotal — direct costs	PHP 250,170,000	73.5%
Project overhead (office, tooling, comms, T&L) — 8%	20,014,000	5.9%
Compliance & audit (ISO 27001, 27701, NPC filings, third-party pen-test)	6,500,000	1.9%
Insurance (professional indemnity, cyber) — 12 mo	2,800,000	0.8%
Bond & performance security financing cost — 5% of ABC × ~1% cost of capital	2,500,000	0.7%
Contingency (7% of direct)	17,512,000	5.1%
Subtotal — burdened cost	PHP 299,496,000	88.0%
Margin (12%)	40,833,000	12.0%
Build price to DepEd, Option A (M1–M8)	PHP 340,329,000	100%

F.9.2 Warranty year — Option A (M9–M20)

Bucket	PHP
Personnel (warranty, \$F.4)	40,180,000
Infrastructure OPEX (12 mo, \$F.5.6 pro-rata)	16,260,000
Software + licensing (year 2, \$F.7 pro-rata)	10,410,000
Overhead (5%)	3,342,500
Contingency (5%)	3,509,600
Margin (10%)	7,370,200
Warranty year price to DepEd, Option A	PHP 81,072,300

F.9.3 Build phase — Option B (cloud, M1–M8, 12 months)

Cloud eliminates CAPEX and reduces some software lines (managed RDS removes DB backup software, managed OS removes RHEL subscription, cloud-native monitoring reduces APM), at the cost of higher monthly OPEX.

Bucket	PHP	% of build
Personnel (build, \$F.4)	102,000,000	42.1%
Cloud OPEX (build 12 mo, \$F.6.1)	47,544,000	19.6%
Software + licensing (reduced for managed services)	7,900,000	3.3%
Training + change mgmt (\$F.8)	20,500,000	8.5%
Subtotal — direct costs	PHP 177,944,000	73.5%
Project overhead (office, tooling, comms, T&L) — 8%	14,236,000	5.9%
Compliance & audit + provider attestation review	7,200,000	3.0%
Insurance (professional indemnity, cyber) — 12 mo	2,800,000	1.2%
Bond & performance security financing cost	2,500,000	1.0%
Contingency (7% of direct)	12,456,000	5.1%
FX hedge / provisional (2% of cloud spend)	951,000	0.4%
Subtotal — burdened cost	PHP 218,087,000	90.1%
Margin (12%)	24,013,000	9.9%
Build price to DepEd, Option B (M1–M8)	PHP 242,100,000	100%

F.9.4 Warranty year — Option B (M9–M20)

Bucket	PHP
Personnel (warranty, §F.4)	40,180,000
Cloud OPEX (12 mo, §F.6.1)	47,544,000
Software + licensing (year 2, reduced)	7,900,000
Overhead (5%)	4,781,000
Contingency (5%)	5,020,000
FX hedge (2% of cloud spend)	951,000
Margin (10%)	10,637,000
Warranty year price to DepEd, Option B	PHP 117,013,000

F.9.5 Total against ABC — both options

	Option A · On-prem	Option B · Cloud	ABC
Build (M1–M8)	340,329,000	242,100,000	—
Warranty year	81,072,300	117,013,000	—
24-month contract price	PHP 421,401,300	PHP 359,113,000	PHP 500,000,000
% of ABC	84.3%	71.8%	100%
Bid headroom	PHP 78,598,700	PHP 140,887,000	—
Headroom %	15.7%	28.2%	—
Year 3 projected run-rate (informational)	~ PHP 65 M	~ PHP 105 M	—

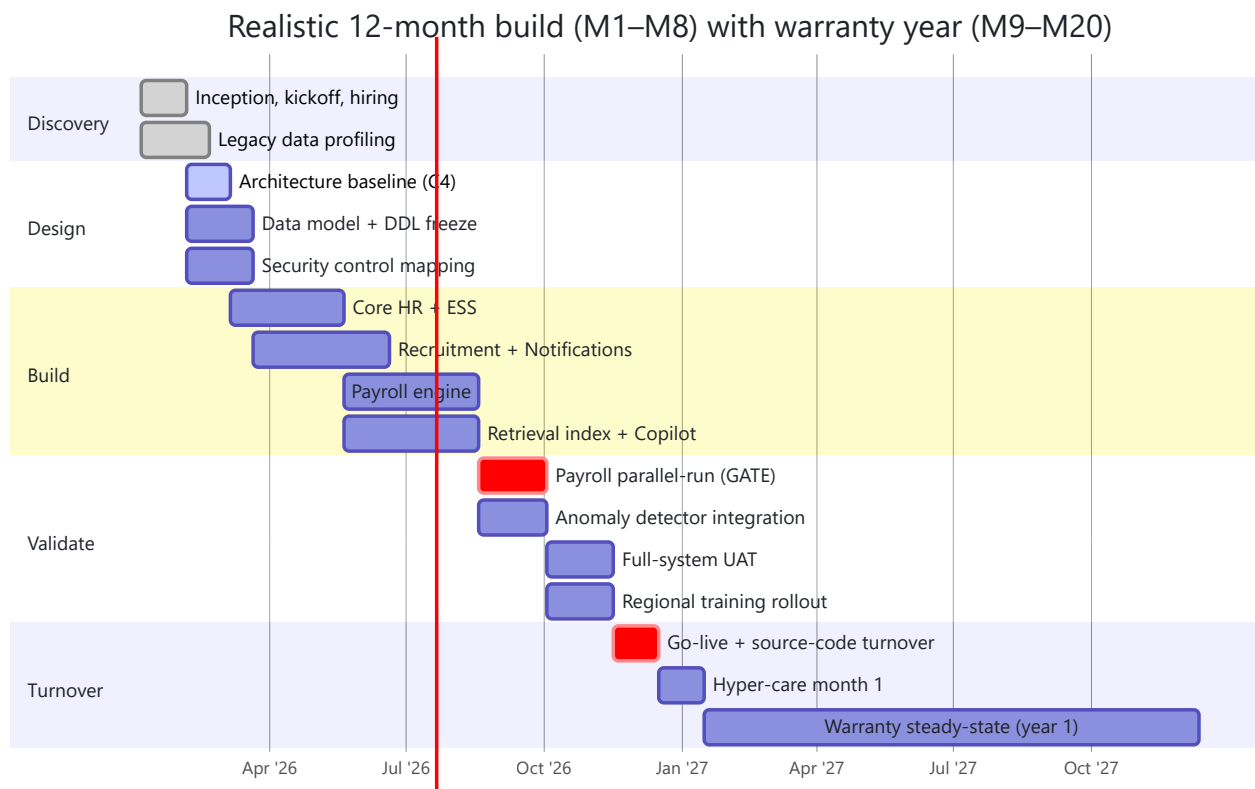
Reading the table

- **Cloud is cheaper over the 24-month contract window** by ~PHP 62 M — no CAPEX, lower software cost, larger headroom for competitive bidding.
- **On-prem is cheaper from year 3 onwards** — hardware amortises to ~PHP 20 M/yr, cloud OPEX stays flat at ~PHP 95 M/yr.
- Both leave meaningful bid headroom for **competitive positioning, additional value-adds, or absorbed contingency**.
- Filipino gov bids typically win at **5–15% below ABC**. That means:
 - Option A supports bids from **PHP 421 M (min) to PHP 475 M (5% below ABC)** with 12% margin intact
 - Option B supports bids from **PHP 359 M (min) to PHP 425 M (5% below ABC)** with substantially wider margin

F.10 Sensitivity analysis — what breaks the model

Scenario	Effect on price	Mitigation
Peso weakens to 60/USD	Option A: +PHP 2–3 M (USD software only) · Option B: +PHP 5–8 M (cloud + software)	FX hedge; GovCloud PH for Option B eliminates most exposure
DR site is required to be a second DepEd-owned facility (not colo)	Option A: +PHP 15–20 M · Option B: n/a	Push back in inception; propose GovCloud PH for DR
Option A chosen but DepEd DC not ready at M2	+PHP 8–12 M (temporary colo bridge)	Add DC-readiness as an M1 gate; fall back to Option B if the gate fails
Option B chosen but data-residency policy tightens	+PHP 4–6 M (all services must pin to GovCloud PH)	Standardise on GovCloud PH from M1 to avoid mid-flight migration
Consulting-firm delivery model (SCC 7 lifted)	+PHP 100–150 M	Not allowed; do not model
Warranty extended to 2 years	+PHP 70–115 M (option A / B respectively)	Price as an add-on; do not bundle into build
User training extended to all 1 M employees in-person	+PHP 40–60 M	Refuse; self-service ESS is the design
PostgreSQL replaced with Oracle DB	+PHP 60–120 M (licenses); larger delta on Option A (no managed alternative)	Refuse; PostgreSQL 16 meets every PBD requirement
Payroll parallel-run extended by 2 cycles	+PHP 6–8 M (personnel + infra)	Absorb in contingency
Adoption at schools slower than projected	+PHP 5–10 M (extended trainer T&L)	Absorb in contingency
Data-privacy incident during build	+PHP 15–25 M	Cyber insurance + rapid-response retainer
Cloud provider price rise mid-contract (Option B)	+PHP 3–6 M	3-yr RI locks pricing; contract clause for major provider changes
On-prem hardware refresh brought forward (Option A)	+PHP 15–20 M (year 5 → year 4)	Warranty capex; DBM continuing appropriation

F.11 Roadmap — month-by-month, with staffing curve



F.11.1 Milestone entry / exit gates (money-flow view)

Milestone	Payment weight	Entry criteria	Exit criteria (billable)
M1 · Inception	10%	NTP received	Project charter, staffing plan, environment plan signed
M2 · Architecture baseline	20%	Inception signed	C4 model, DDL, security controls, SLA design approved
M3 · Core HR + ESS	15%	Baseline signed	Modules 1, 4, 5 in SIT; > 90% test-pass rate
M4 · Recruitment	10%	Core HR delivered	Modules 2, 3 in SIT with 28 report views generated
M5 · Payroll build	15%	Core HR data live	Payroll engine + retrieval index in SIT; anomaly detector trained
M6 · Payroll parallel-run gate	10%	Payroll in UAT	3 consecutive payroll cycles matched to legacy within tolerance
M7 · UAT + training	10%	M6 passed	200 UAT scripts passed; 35 K users trained; go-live sign-off
M8 · Final acceptance	10%	Go-live stable 30 days	Source-code turnover, warranty starts, formal handover

F.11.2 Warranty year — service levels

Year 2 runs at the **P1–P4 SLAs from PBD §7** (see [Paper A §A.3](#)):

Severity	Response	Resolution	Notes
P1 Critical	1 h	24 h	Payroll blocked, auth broken, data loss
P2 High	2 h	48 h	Module unavailable, integration down
P3 Medium	4 h	5 working days	Feature bug, single-user impact
P4 Low	1 working day	Next release	Enhancement, cosmetic

The warranty team (§F.3.3) is sized to meet P1/P2 with a **follow-the-sun rotation across two shifts** (07:00–19:00 + 19:00–07:00), covering PBD’s 24×7 P1 requirement without a 3-shift roster (personnel cost would rise ~1.7×).

F.12 What we deliberately do not include

Line items **not** in this model, and why:

1. **Legacy data migration beyond initial load.** One-off migration at M3 is included; on-going reconciliation with legacy PIS after go-live is a separate T&M engagement.
2. **Integration with any regulator not named in the PBD.** GSIS, Pag-IBIG, PhilHealth, SSS, BIR, CSC, DBM, NPC, DICT are in scope. Any additional agency (e.g. LandBank beyond the disbursement API, DFA, PRC) is a change order.
3. **A native mobile app.** The design is a Progressive Web App (Paper D §2). A native iOS/Android app duplicates effort; if DepEd insists, add PHP 20–35 M.
4. **On-site presence at 200,000+ schools.** Field-level support is via SDO liaison officers, not the prime contractor. Prime contractor coverage is Central Office + 17 Regional Offices only.
5. **Any AI / ML component beyond the Copilot (D §3) and anomaly detector (D §4).** Both are included in the personnel + infra lines; nothing else assumes GPU capacity or third-party foundation models.
6. **Custom hardware for schools.** Kiosks, thin clients, and tablets remain a separate DepEd procurement.
7. **Post-warranty maintenance.** Years 3+ are a separate MSA — typically priced at 15–20% of build price per year for public-sector systems at this scale.

F.13 Comparables — is PHP 500 M realistic?

Cross-checked against public procurement data for comparable-scale gov HRIS:

System	Country	Scale	Reported build cost	Duration	PHP-equivalent
Novopay (rebuilt)	NZ	110 K teachers	NZD 45 M+ (rebuild only)	3 yr rebuild	~ PHP 1.5 B
HRMIS	Malaysia	1.6 M civil servants	RM 200 M (est.)	~ 5 yr	~ PHP 2.6 B
Shala Darpan + UDISE+	India	9 M teachers	INR 400 Cr (approx.)	Multi-phase	~ PHP 2.7 B
SIASN / SAPK	Indonesia	4.2 M ASN	IDR 800 B (est.)	~ 4 yr	~ PHP 2.8 B
DepEd HRIS (this model)	PH	1 M+ employees	PHP 421 M	12 mo build + 12 mo warranty	—

The PBD ABC is priced at **~15–30% of the fully-mature cost of comparable systems**. This is consistent with a **first-generation delivery** followed by iterative hardening — the same pattern every gov HRIS in the reference set actually followed. It is realistic **if** DepEd budgets for years 3+ under a separate MSA and does not expect year-1 to reach the fit-and-finish of NHS ESR or HRMIS.

Bottom line: PHP 500 M is realistic for a solid first-generation delivery. It is not enough for a mature system in year 1 — nobody's is.